

TRISTAN TOSHIHARU ENDO

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EDUCATION

University of California, San Diego

Expected June 2027

B.S. Math-Computer Science

La Jolla, CA

- **Relevant Coursework:** Abstract Algebra, Brownian Motion, Combinatorics, Probability Theory, Quantitative Finance, Real Analysis, Statistics, Stochastic Processes

EXPERIENCE

Tekly Studio, Quantitative Research & AI Engineering Lab

Jan 2026 – August 2026

Quantitative Researcher / AI Engineering Intern

San Francisco, CA

- Built and backtested ML-driven trading strategies in Python (scikit-learn, pandas) on financial time-series data, evaluating performance via Sharpe ratio and drawdown analysis
- Engineered data pipelines for preprocessing, simulation, and model evaluation, enabling rapid iteration across strategy variants

Think Twice Legal

Apr 2025 – July 2026

Audio Engineer / Analyst

San Bruno, CA

- Optimized Demucs-based source separation models via hyperparameter tuning, improving reconstruction fidelity for forensic audio analysis used in legal proceedings
- Built preprocessing and normalization pipelines for consistent model input across varying sample rates and noise conditions

RESEARCH

UC San Diego, Undergraduate Researcher

Jan 2026 – Mar 2026

Advisor: Alexander Cloninger

La Jolla, CA

- **Neural Network Gaussian Processes:** Derived closed-form conjugate kernels for ReLU/erf networks in Python; validated finite-width network simulations against infinite-width theoretical predictions

UC San Diego, Undergraduate Researcher

Feb 2026 – Present

Advisor: Raymond O. Chou

La Jolla, CA

- **Springer Fibers in Algebraic Geometry:** Built a Python/SymPy toolchain enumerating Schubert cells with their Plücker coordinates and deriving cell-closure relations via Gröbner-basis elimination; computationally verified cell-poset invariants (irreducible components, Betti numbers)

SELECTED PROJECTS

Springer Fiber Visualizer | *Python, JavaScript, HTML Canvas*

Feb 2026 – Present

- Built a 3D web app rendering Springer-fiber cells as spheres with directional closure arrows and interactive exploration; animated cell degenerations as nested “sphere-within-sphere” limits that track the surviving free parameters

Audio Signal Processing & ML Pipeline | *Python, Demucs*

Apr 2025 – July 2026

- Designed FFT-based noise reduction and stem-separation pipelines for isolating target signals in complex audio environments; built normalization workflows for consistent model input across varied audio conditions

Stochastic Simulation of Brownian Motion & OU Processes | *Python*

Sep 2025 – Dec 2025

- Simulated fractional Brownian motion across varying Hurst parameters to characterize long-range dependence; quantified empirical extrema distributions via large-scale Monte Carlo simulation

TECHNICAL SKILLS

Languages: Python, C, C++, Java, SQL, R, MATLAB, HTML

ML & Data: PyTorch, scikit-learn, pandas, NumPy, SymPy, FFT/signal processing

Tools: Git, Linux, Jupyter, L^AT_EX

Domains: Algorithmic trading, time-series modeling, stochastic processes, audio source separation, statistical modeling, computational algebraic geometry